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(54) **SEMICONDUCTOR DEVICE AND METHOD OF MANUFACTURING THE SAME**

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(52) **U.S. Cl.**
CPC ***H10B 12/09*** (2023.02); ***H10B 12/315*** (2023.02); ***H10B 12/50*** (2023.02)

(58) **Field of Classification Search**
CPC .. H10B 12/033; H10B 12/0335; H10B 12/00; H10B 12/31; H01L 28/91; H01L 23/5223; H01L 28/90
See application file for complete search history.

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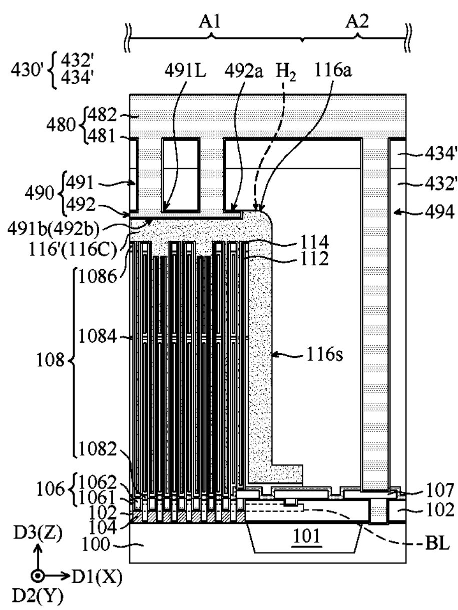
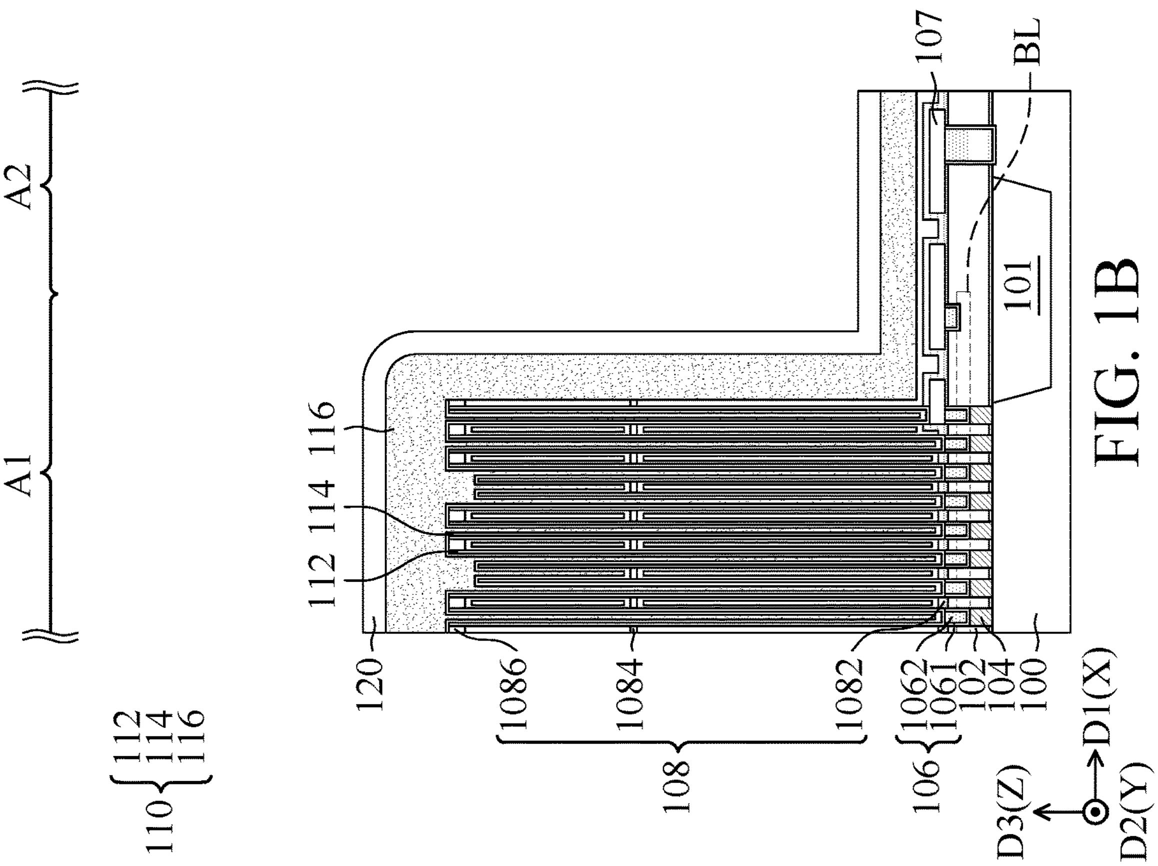
(57) **ABSTRACT**

A semiconductor device includes a first insulating layer over a substrate and a contact plug in the first insulating layer and in contact with the surface of the substrate. The semiconductor device further includes a capacitor structure above the contact plug and a second insulating layer on the first insulating layer and covering the capacitor structure. The capacitor structure includes a conductive layer over the first insulating layer. The semiconductor device further includes a capacitor contact over the capacitor structure. The capacitor contact includes a first contact portion and a second contact portion. The first contact portion penetrates through the second insulating layer and is in contact with the conductive layer of the capacitor structure. The second contact portion connects the outer surface of the first contact portion, and surrounds the lower portion of the first contact portion.

20 Claims, 9 Drawing Sheets

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1084
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108
1062
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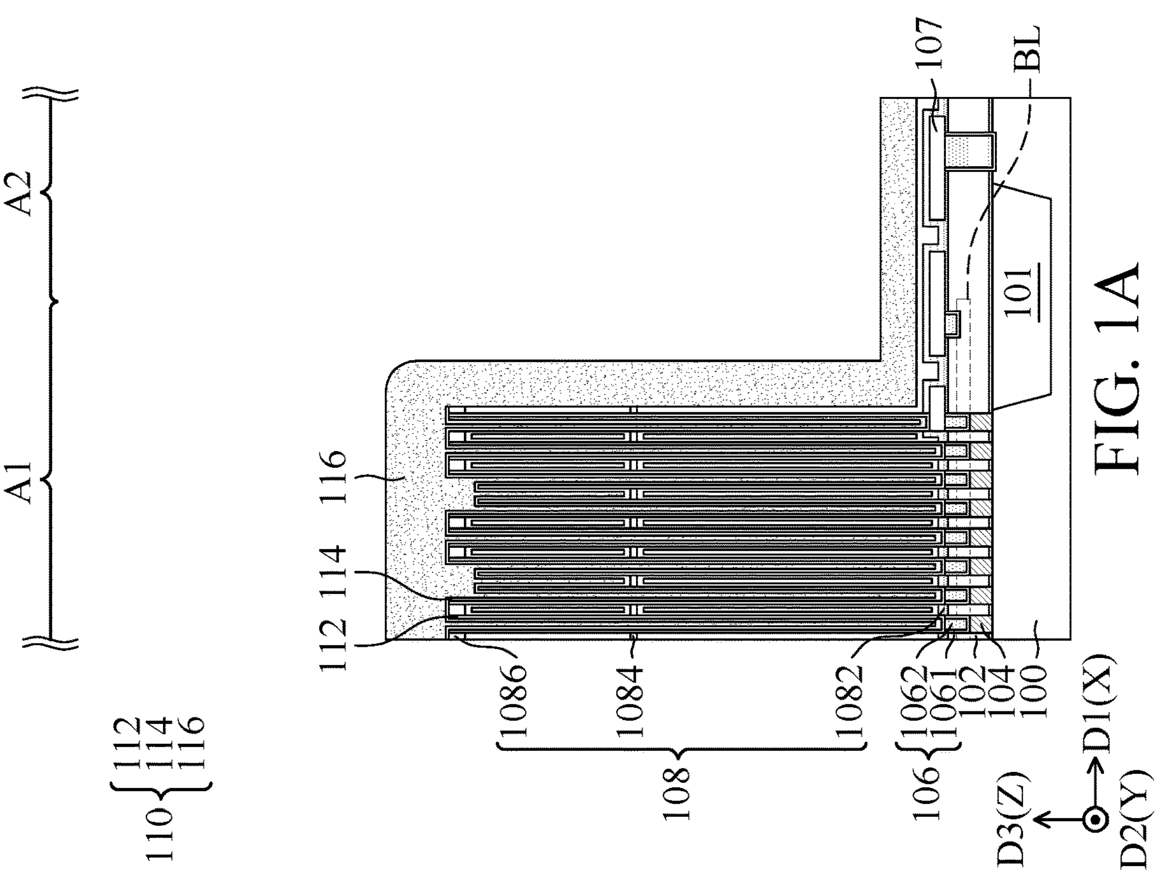
106
102
104
100
107
BL
101
D3(Z)
D1(X)
D2(Y)

FIG. 1B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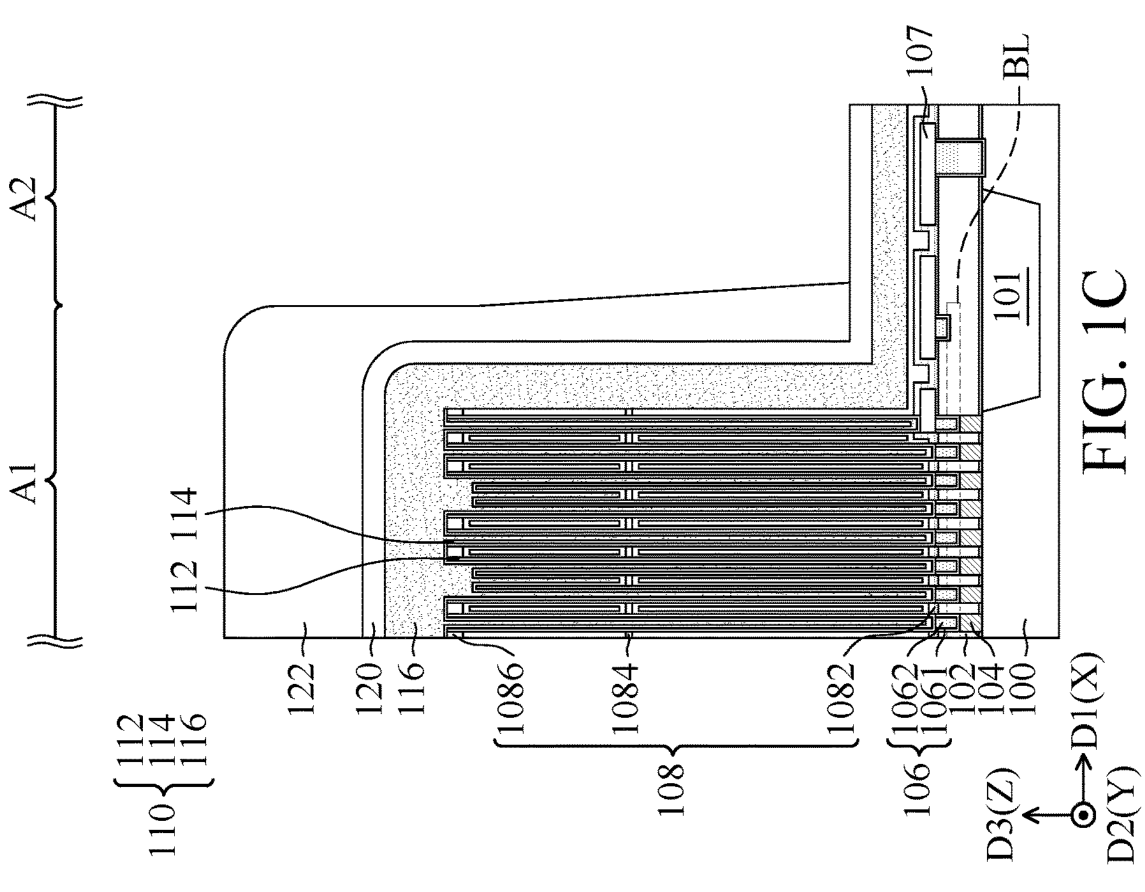
101
D3(Z)
D1(X)
D2(Y)

FIG. 1A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114
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122
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1086
1084
1082
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1061
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104
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108
106
107
BL
101
D3(Z)
D1(X)
D2(Y)

FIG. 1C

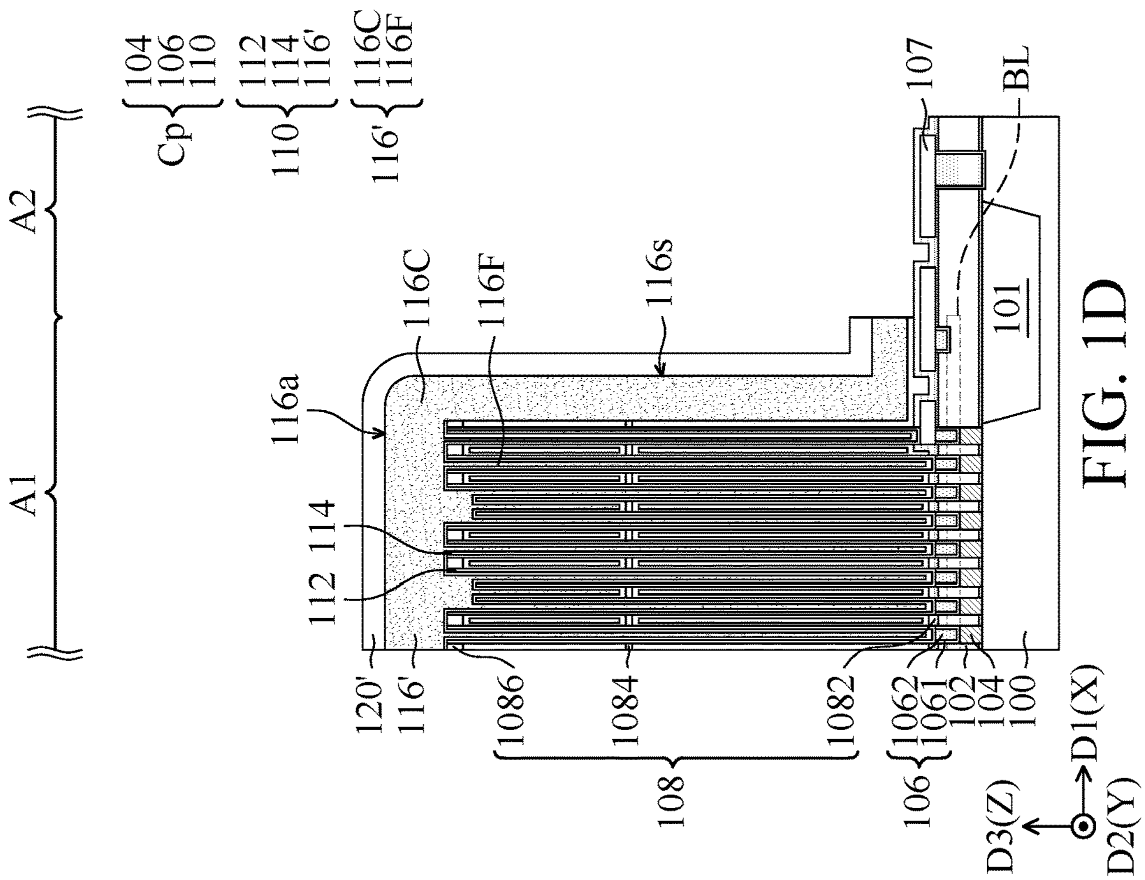

A1
A2
Cp
104
106
110
110
112
114
116'
116'
116C
116F
116a
116s
120'
116'
1086
1084
1082
1062
1061
102
104
100
108
106
107
BL
101
D3(Z)
D1(X)
D2(Y)

FIG. 1D

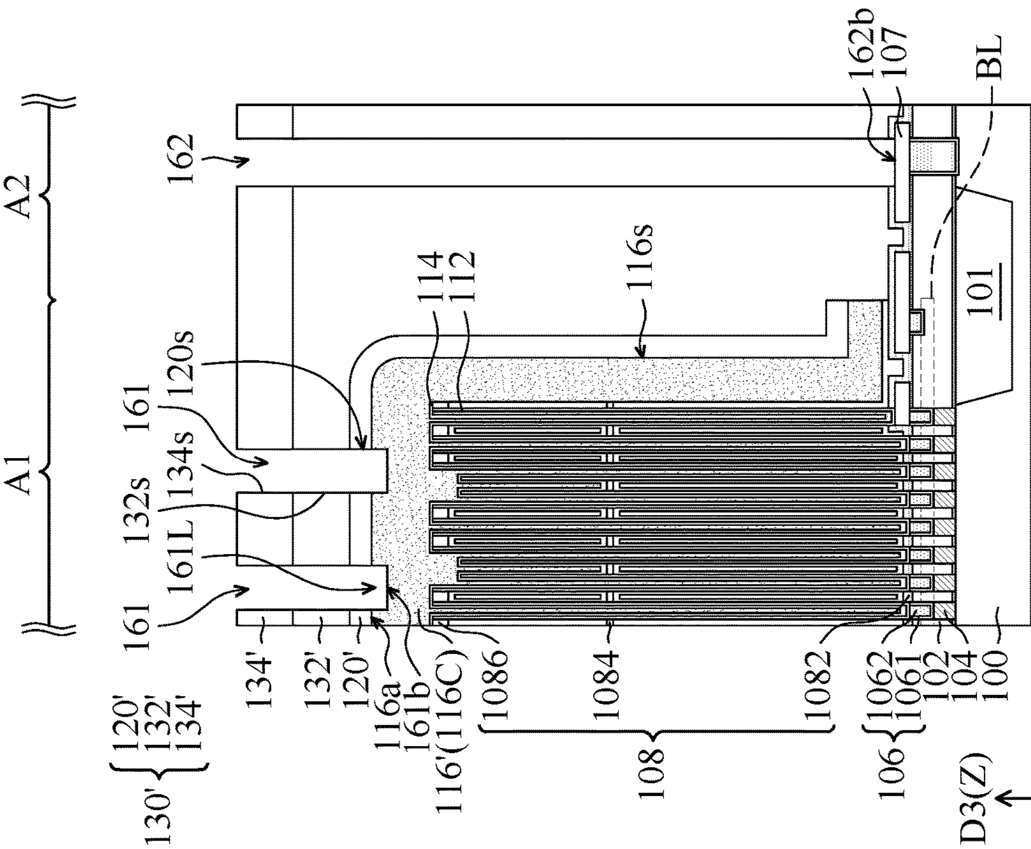
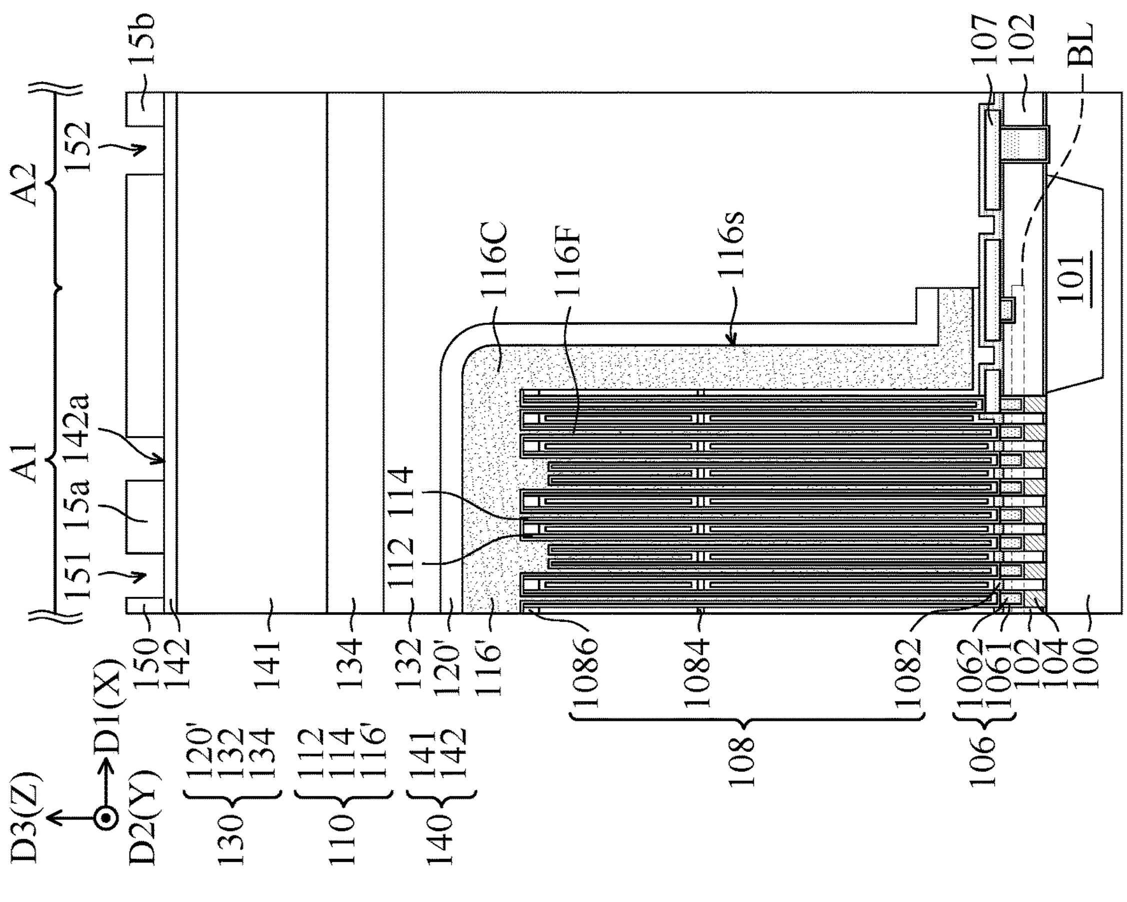

A1
A2
161
161
161L
132s
134s
120s
162
162b
107
BL
116s
114
112
101
134'
132'
120'
116a
161b
116'(116C)
1086
1084
1082
1062
1061
102
104
100
130'
120'
132'
134'
108
106
D3(Z)
D1(X)
D2(Y)

FIG. 1F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D1(X)
D2(Y)
130
120'
132
134
110
112
114
116'
140
141
142
A1
A2
151
15a
142a
152
15b
150
142
141
134
132
120'
116'
112
114
116C
116F
116s
107
102
BL
101
1086
1084
1082
1062
1061
102
104
100
108
106

FIG. 1E

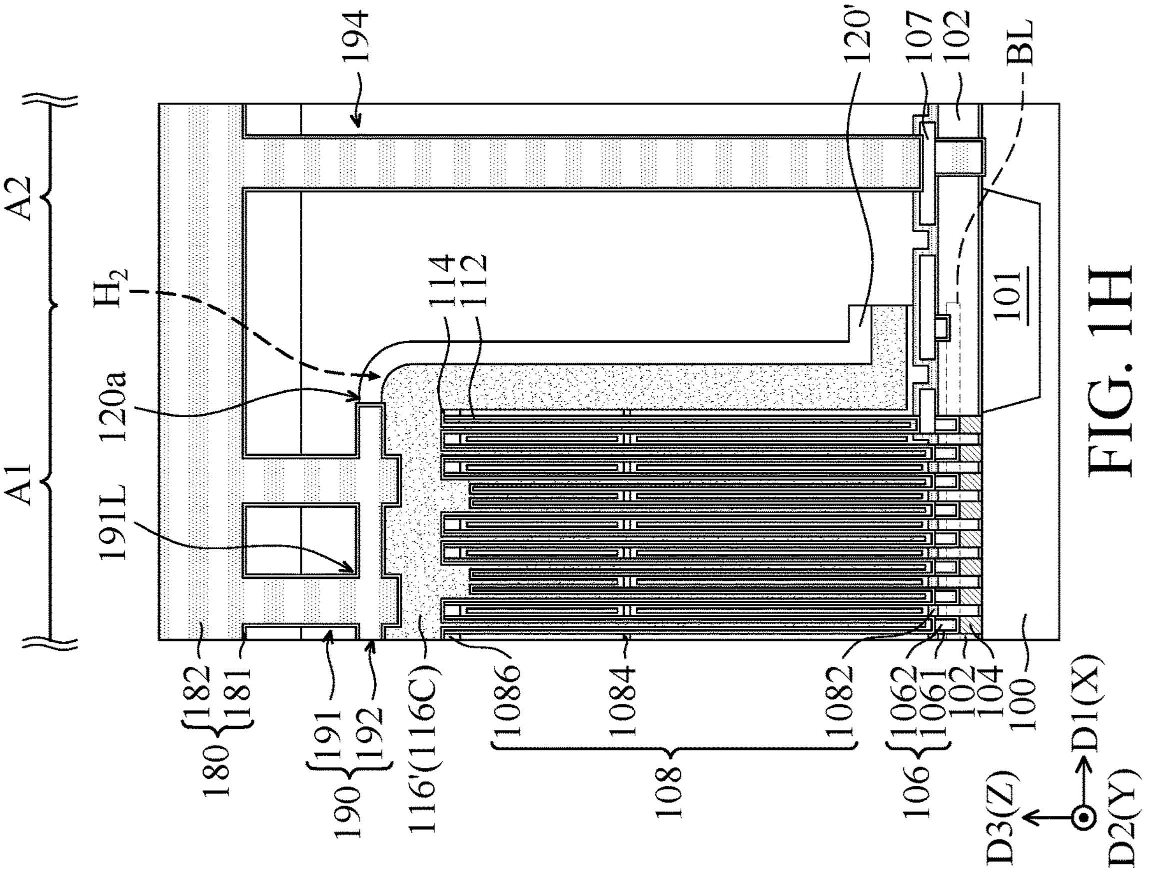

A1
A2
194
H$_2$
120a
191L
120'
114
112
107
102
BL
101
182
181
180
191
192
190
116'(116C)
1086
1084
1082
108
1062
1061
106
102
104
100
D3(Z)
D1(X)
D2(Y)

FIG. 1H

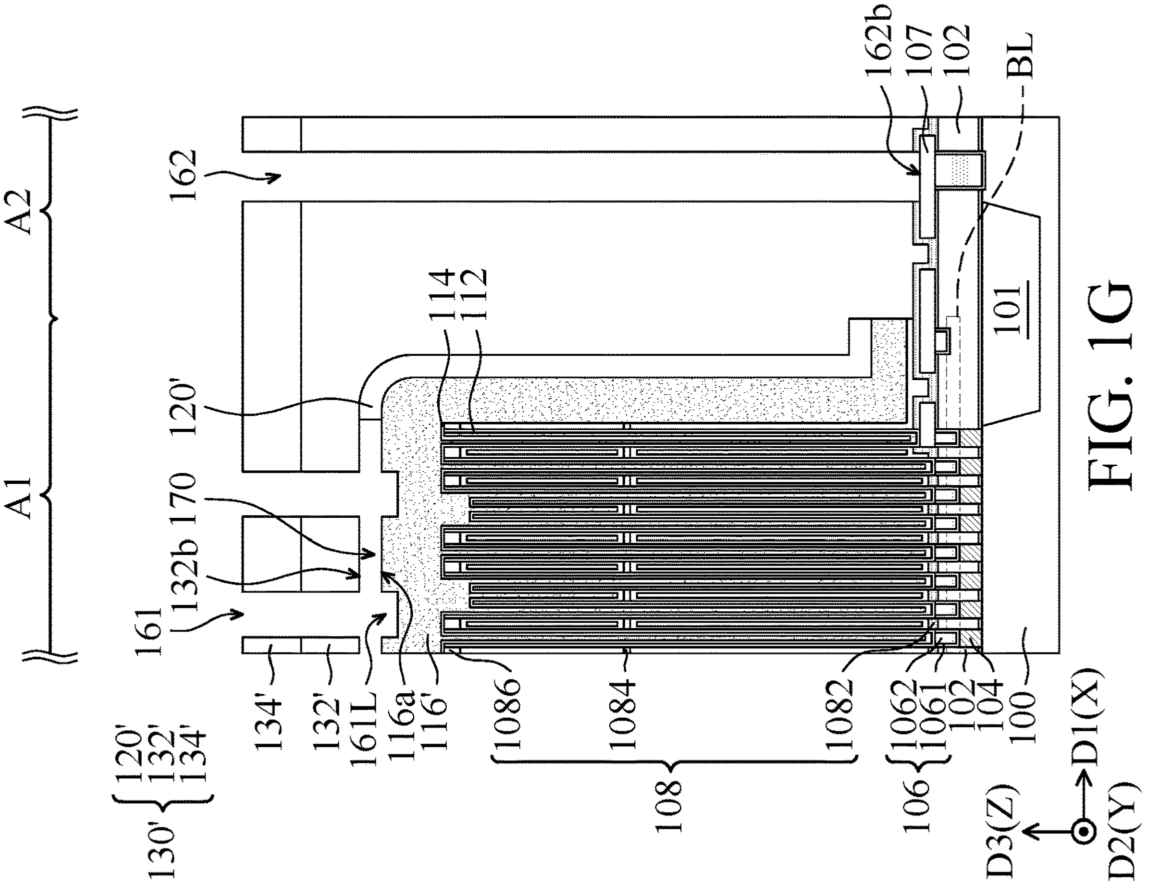

A1
A2
161
132b
170
120'
162
114
112
162b
107
102
BL
101
120'
132'
134'
130'
134'
132'
161L
116a
116'
1086
1084
1082
108
1062
1061
106
102
104
100
D3(Z)
D1(X)
D2(Y)

FIG. 1G

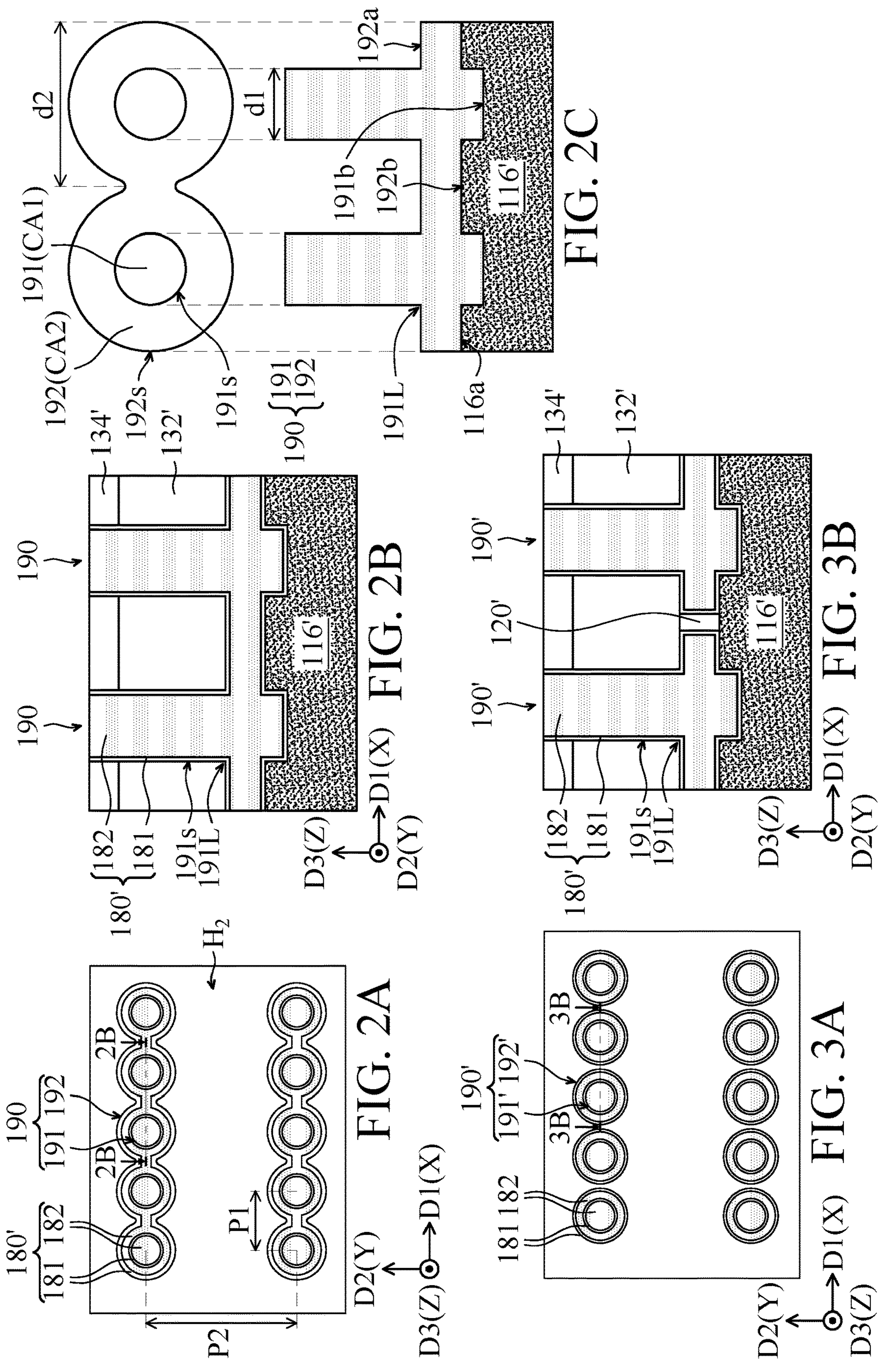


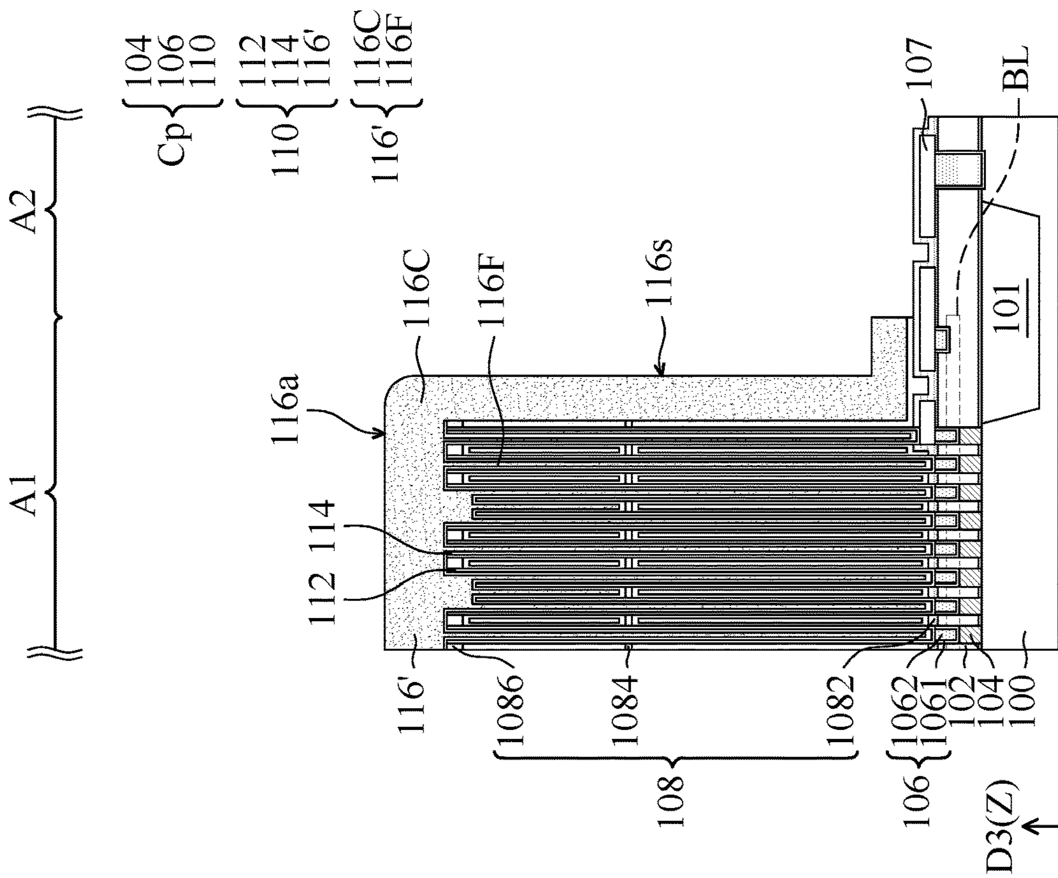
FIG. 2A

FIG. 2B

FIG. 2C

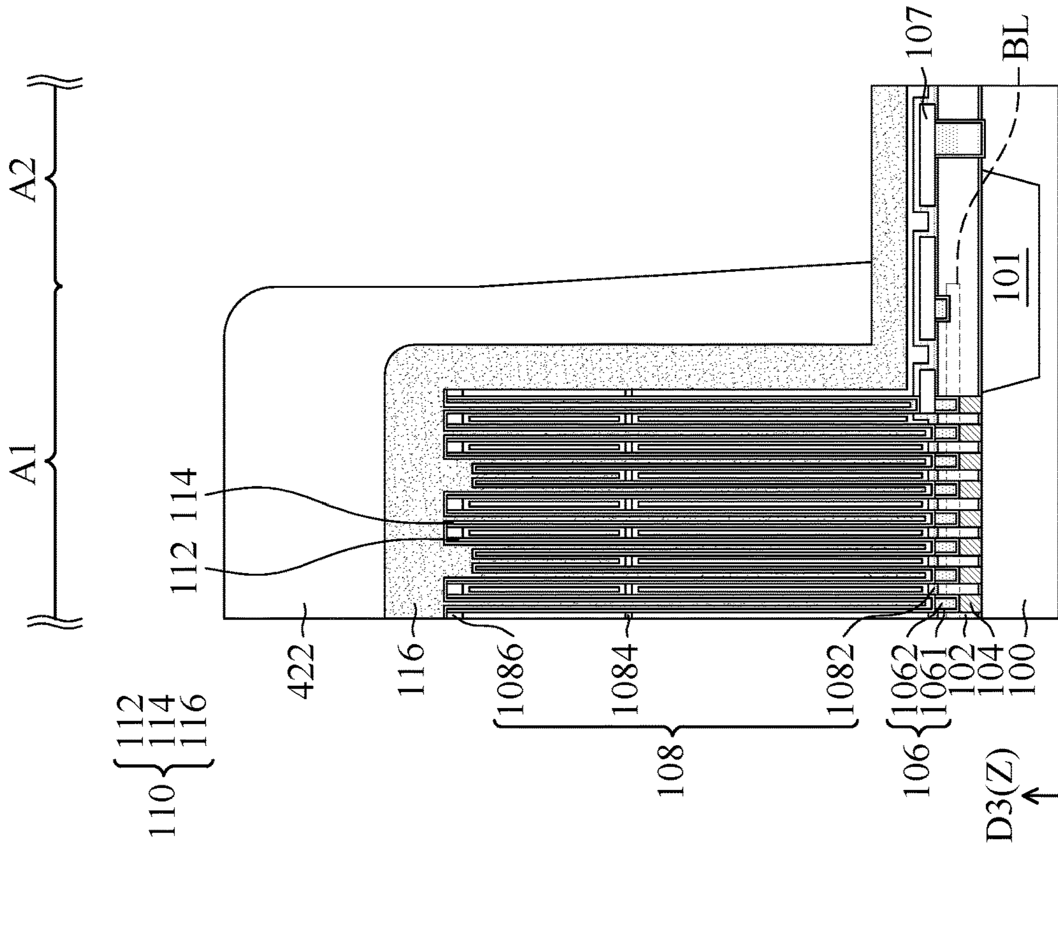
FIG. 3A

FIG. 3B



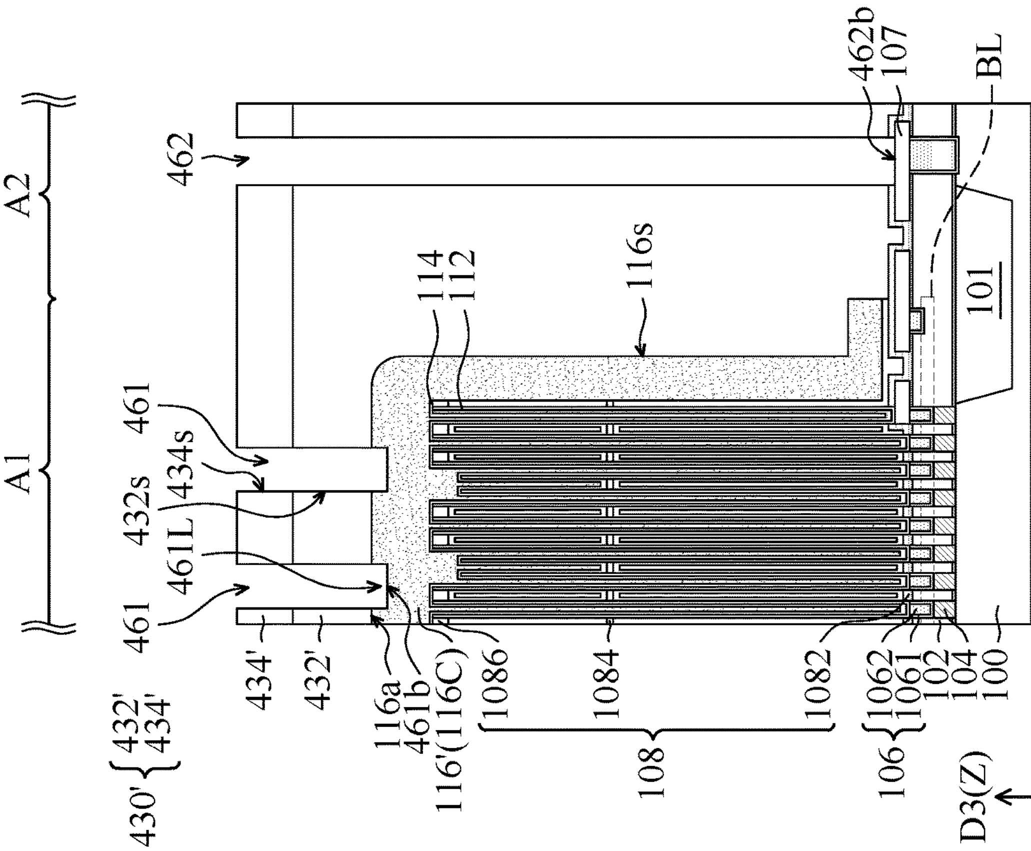
A1
A2
Cp{104 106 110
110{112 114 116'
116'{116C 116F
116a
116C
116F
116s
112 114
107
BL
101
116'
1086
1084
1082
108
1062
1061
106
102
104
100
D3(Z)
D2(Y)
D1(X)

FIG. 4B



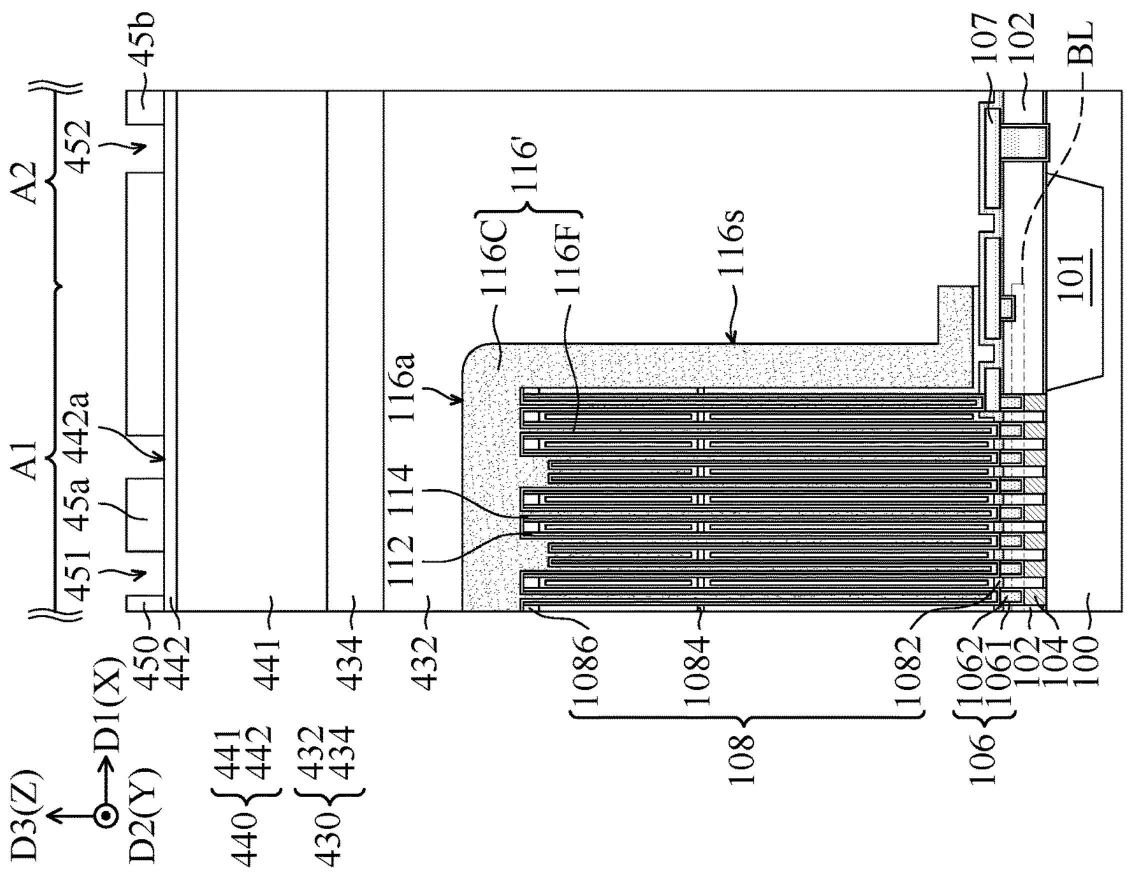
A1
A2
110{112 114 116
112 114
107
BL
101
422
116
1086
1084
1082
108
1062
1061
106
102
104
100
D3(Z)
D2(Y)
D1(X)

FIG. 4A



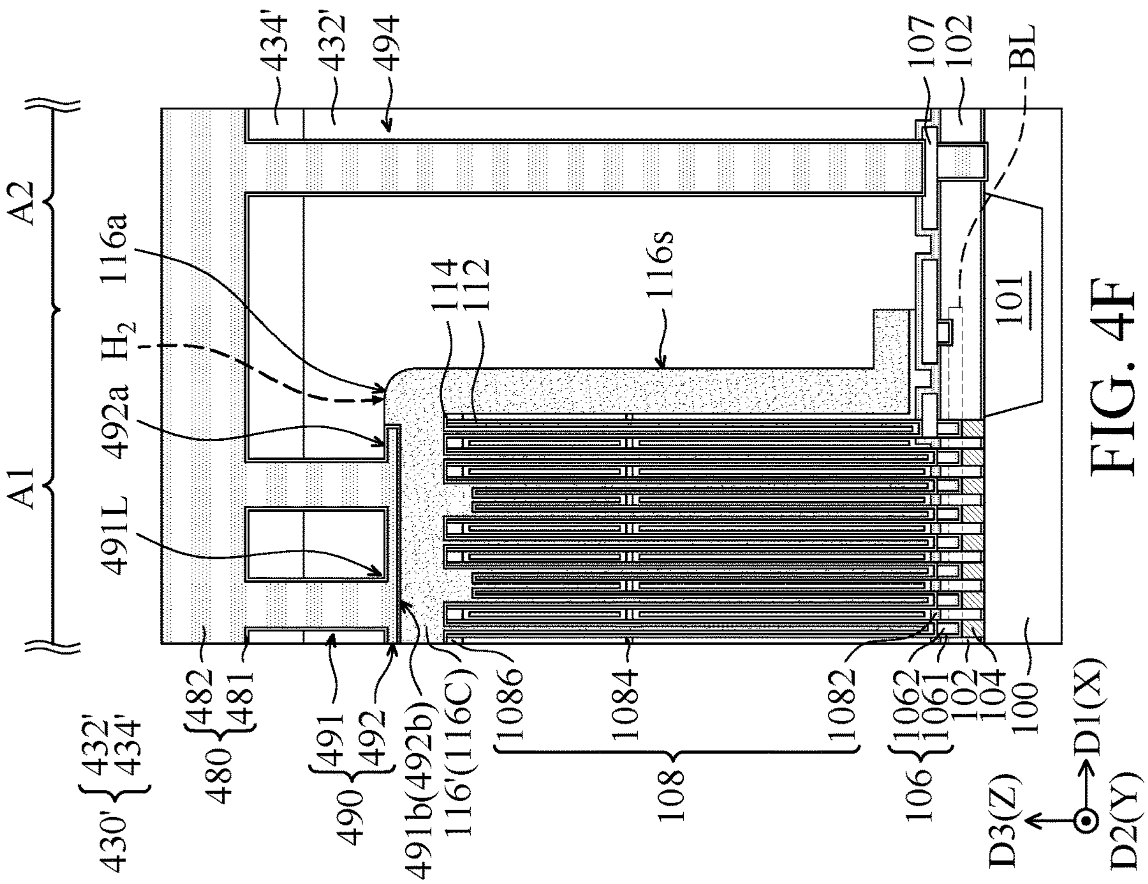
A1
A2
461
461L
432s
434s
462
462b
107
BL
116s
114
112
101
434'
432'
116a
461b
116'(116C)
1086
1084
1082
1062
1061
102
104
100
430'
108
106
D3(Z)
D1(X)
D2(Y)

FIG. 4D



A1
A2
451
45a
442a
452
45b
116C
116F
116'
116s
116a
112
114
107
102
BL
101
450
442
441
434
432
1086
1084
1082
1062
1061
104
100
440
430
108
106
D3(Z)
D1(X)
D2(Y)

FIG. 4C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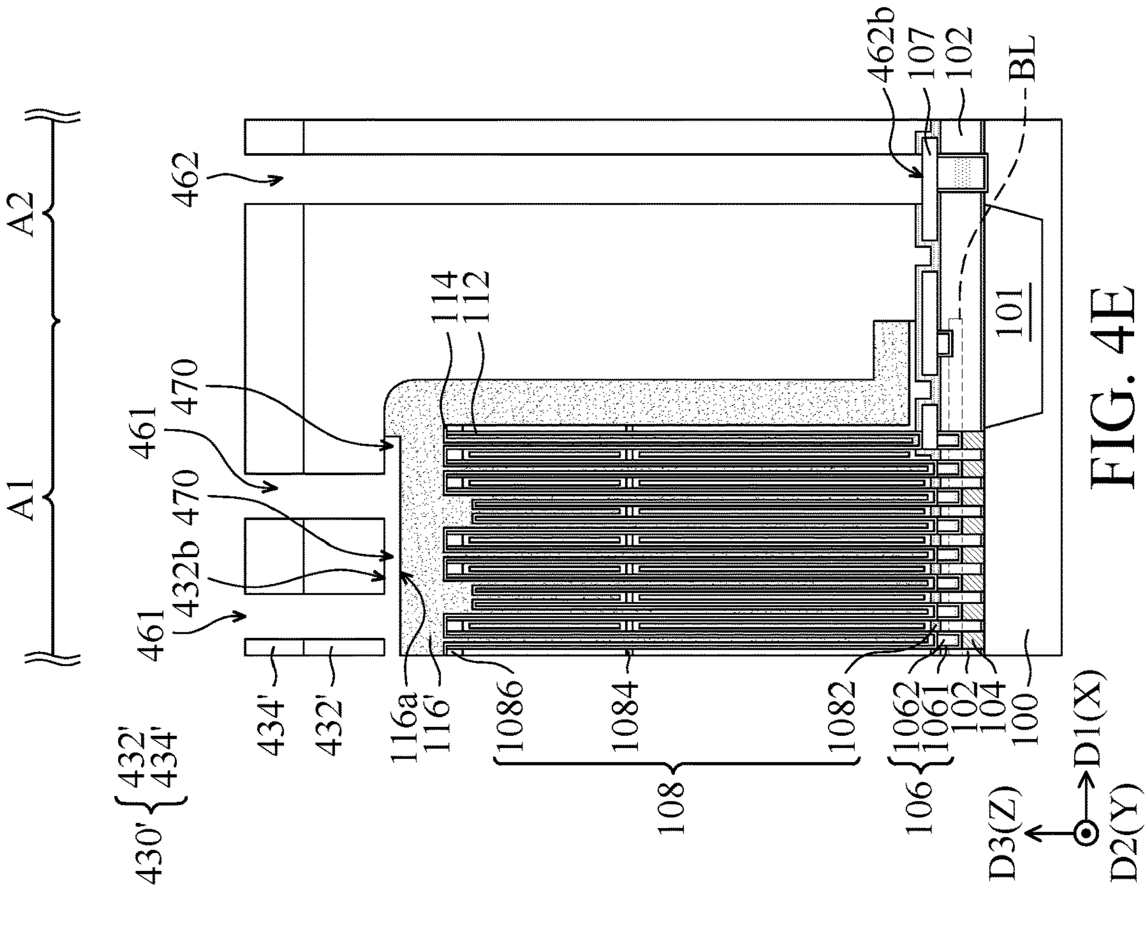
A2
491L
492a
H_2
116a
116s
114
112
434'
432'
494
107
102
BL
101
430'
432'
434'
480
482
481
490
491
492
491b(492b)
116'(116C)
1086
1084
1082
108
106
1062
1061
102
104
100
D3(Z)
D2(Y)
D1(X)

FIG. 4F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A2
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432b
470
461
470
462
114
112
462b
107
102
BL
101
430'
432'
434'
434'
432'
116a
116'
1086
1084
1082
108
106
1062
1061
102
104
100
D3(Z)
D2(Y)
D1(X)

FIG. 4E

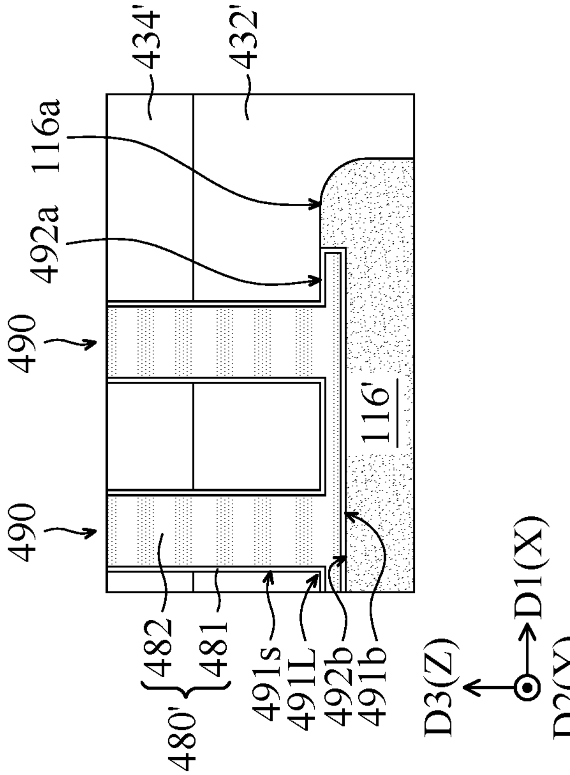
434'
432'
116a
492a
490
490
116'
482
481
480'
491s
491L
492b
491b
D3(Z)
D1(X)
D2(Y)

FIG. 5B

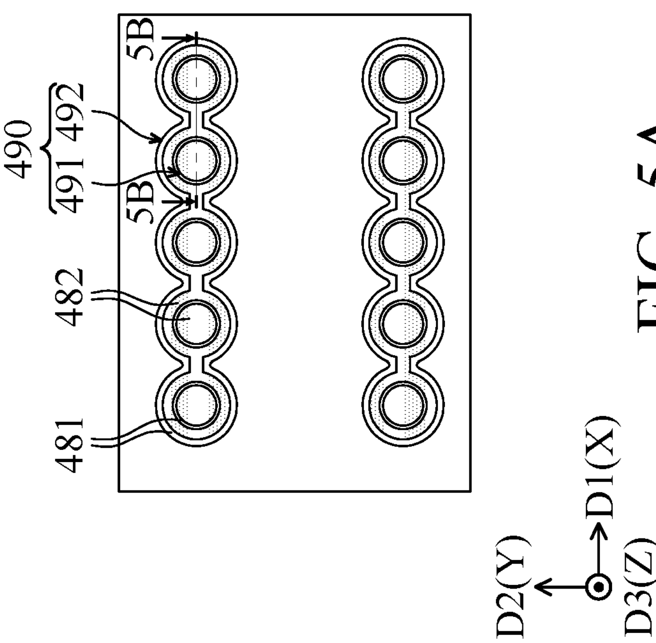
490
491
492
5B
5B
482
481
D2(Y)
D1(X)
D3(Z)

FIG. 5A

SEMICONDUCTOR DEVICE AND METHOD OF MANUFACTURING THE SAME

BACKGROUND

Technical Field

The disclosure relates to a semiconductor device and method of manufacturing the same, and it relates to a semiconductor device that includes a capacitor contact and method of manufacturing the same.

Description of the Related Art

Dynamic random access memory (DRAM) technology is widely used in consumer electronic products. In order to increase the integration density of components in DRAM and improve overall electrical performance, the current techniques used in the fabrication of DRAM continues to trend toward scaling down the components. As the components continue to shrink, many challenges arise. For example, a memory cell plate in an array region of a conventional semiconductor device may be covered by a metal layer, which reduces the resistance of a contact that is subsequently formed on the metal layer. However, the metal layer blocks the flow paths of the gas that is required to repair the process defects of the memory cell plate in the back-end-of-line (BEOL) process.

SUMMARY

Some embodiments of the present disclosure provide semiconductor devices includes a first insulating layer over a substrate, a contact plug in the first insulating layer and in contact with a surface of the substrate, a capacitor structure above the contact plug, a second insulating layer over the first insulating layer and a capacitor contact over the capacitor structure. The capacitor structure includes a conductive layer over the first insulating layer. The second insulating layer covers the capacitor structure. The capacitor contact includes a first contact portion and a second contact portion. The first contact portion penetrates through the second insulating layer and is in contact with the conductive layer of the capacitor structure. The second contact portion connects the outer surface of the first contact portion and surrounds the lower portion of the first contact portion.

Some embodiments of the present disclosure provide methods of manufacturing a semiconductor device includes providing a substrate and forming a memory cell plate over the substrate. The memory cell plate includes a first insulating layer over the substrate, a contact plug in the first insulating layer and a capacitor structure above the contact plug. The contact plug is in contact with a surface of the substrate. The capacitor structure includes a conductive layer over the first insulating layer. The method of manufacturing the semiconductor device further includes forming a second insulating layer over the first insulating layer, and the second insulating layer covers the memory cell plate. The method of manufacturing the semiconductor device further includes forming a capacitor contact over the capacitor structure. The capacitor contact includes a first contact portion and a second contact portion. The first contact portion penetrates through the second insulating layer and is in contact with the conductive layer of the capacitor structure. The second contact portion connects the outer surface of the first contact portion and surrounds the lower portion of the first contact portion.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A, FIG. **1**B, FIG. **1**C, FIG. **1**D, FIG. **1**E, FIG. **1**F, FIG. **1**G and FIG. **1**H are cross-sectional views of various fabrication stages of a semiconductor device, in accordance with some embodiments of the present disclosure.

FIG. **2**A is a schematic top view of the capacitor contacts in the array region, in accordance with some embodiments of the present disclosure.

FIG. **2**B is a cross-sectional view taken along sectional line **2**B-**2**B of the structure including capacitor contacts of FIG. **2**A.

FIG. **2**C is a schematic diagram of a first contact portion and a second contact portion of each of the capacitor contacts, in accordance with some embodiments of the present disclosure.

FIG. **3**A is a schematic top view of the capacitor contacts in the array region, in accordance with some other embodiments of the present disclosure.

FIG. **3**B is a cross-sectional view taken along sectional line **3**B-**3**B of the structure including capacitor contacts of FIG. **3**A.

FIG. **4**A, FIG. **4**B, FIG. **4**C, FIG. **4**D, FIG. **4**E and FIG. **4**F are cross-sectional views of various fabrication stages of a semiconductor device, in accordance with some embodiments of the present disclosure.

FIG. **5**A is a schematic top view of the capacitor contacts in the array region, in accordance with some embodiments of the present disclosure.

FIG. **5**B is a cross-sectional view taken along sectional line **5**B-**5**B of the structure including capacitor contacts of FIG. **5**A.

DETAILED DESCRIPTION

According to a semiconductor device and method of manufacturing the semiconductor device of the embodiments, a contact area between a conductive layer of a capacitor structure and a capacitor contact can be increased, so as to reduce the high resistance of the capacitor contact due to the material of the conductive layer (for example, silicon germanium material). In addition, when a hydrogen sintering process is performed, hydrogen is able to successfully passivate silicon dangling bonds at the Si-containing surface of the conductive layer of the capacitor structure, in accordance with the present disclosure. Therefore, the component defects of the semiconductor device can be repaired, so as to reduce the threshold voltage of the semiconductor device and facilitate the high-frequency operation and application of the semiconductor device.

In addition, the semiconductor device and the manufacturing method thereof are, for example, applied to a DRAM. Although the cross-sectional views of the semiconductor device, in accordance with some embodiments only show a portion of an array region (in which a portion of a memory cell plate is formed) and a portion of a peripheral region adjacent to the array region for the illustration, the present disclosure is not limited to those illustrated components.

Referring to FIG. **1**A, a substrate **100** includes an array region **A1** and a peripheral region **A2** are provided. The substrate **100** may include a semiconductor material, such as silicon, gallium arsenide, gallium nitride, germanium silicide, another suitable material, or a combination thereof. In one embodiment, the substrate **100** is a silicon-on-insulator (SOI) substrate.

Various components may be formed in the substrate **100**; for example, buried word lines (not shown), isolation structures (such as an isolation structure **101** that separates the array region A**1** from the peripheral region A**2**, and another isolation structure (not shown) that separates the active regions in the array region A**1**) and bit lines BL (positioned behind the barrier structures **106** but not on the current cross section, drawn in dashed line). In addition, various components may be formed above the substrate **100**. For example, contact plugs that contact active regions of the substrate **100**, capacitor structures disposed on the contact plugs, and interconnection structures in the BEOL process. Some of those components that are in or above the substrate **100** are omitted in the diagrams for the sake of simplicity and clarity.

Referring to FIG. **1**A, a first insulating layer **102** is formed above the substrate **100**. Several contact plugs **104** formed in the first insulating layer **102** in the array region A**1**. The contact plugs **104** are in contact with the surface of the substrate **100** and active regions (not shown) of the substrate **100**, and the contact plugs **104** corresponding to active regions (not shown). The first insulating layer **102** may be a single-layer structure or a multilayer structure. For example, the first insulating layer **102** is a double-layer structure that includes an oxide layer and a nitride layer on the oxide layer.

The contact plug **104** may be a multilayer structure. In some embodiments, the contact plug **104** includes a non-metallic conductive element, a conductive liner on the non-metallic conductive element, and a metallic conductive element on the conductive liner. Each of the contact plugs **104** is configured as a single-layer structure is shown in figures for the purpose of simplicity and clarity.

In addition, several barrier structures **106** are formed above the contact plugs **104** and in contact with the associated contact plugs **104**. Subsequently, a capacitor structure is formed on each of the barrier structures **106**. The area of the bottom surface of the barrier structures **106** may be equal to or greater than the area of the top surface of the contact plug **104**. Accordingly, the top surface of the contact plug **104** can be completely covered by the bottom surface of the barrier structure **106** above the associated contact plug **104**. Therefore, the barrier structures **106** can prevent the etchant used in the etching process from infiltrating and damaging the contact plugs **104**.

The barrier structures **106** include a first barrier layer **1061** and a second barrier layer **1062**. The first barrier layer **1061** covers the sidewalls and the bottom surface of the second barrier layer **1062**. The first barrier layer **1061** can be made of a material with good adhesion with the first insulating layer **102**. Also, the material of the first barrier layer **1061** may have good adhesion with the electrodes (such as the first electrodes **112**) of the capacitor structures to be formed subsequently, so as to prevent an etchant used in the etching process form penetrating into the barrier structures **106** through the electrodes of the capacitor structures. In addition, the material of the first barrier layer **1061** can prevent the etchant from infiltrating into the substrate **100** through the gaps between the barrier structures **106** and the first insulating layer **102**, thereby preventing the components in the substrate **100** from being damaged. The first barrier layer **1061** may include titanium, titanium nitride, tungsten nitride, tantalum, tantalum nitride, another suitable material, or a combination thereof. The second barrier layer **1062** may include tungsten, copper, another suitable metal materials with good electrical conductivity, or a combination thereof to provide a lower electrical resistance value.

As shown in FIG. **1**A, capacitor structures **110** are formed above the contact plugs **104** in the array region A**1**. Specifically, the capacitor structures **110** are in contact with and disposed on the barrier structures **106**. In some embodiments, the capacitor structures **110** extend in the third direction D**3** (e.g. the Z-direction).

The capacitor structures **110** include first electrodes **112**, a dielectric layer **114**, and a conductive layer **116'** (also referred to as a second electrode) (FIG. **1**D). A conductive material **116** is patterned to form the conductive layer **116'**. The first electrodes **112** include, for example, several independent first electrode material elements each having a U-shaped cross section. The first electrodes **112** may include titanium, titanium nitride, tantalum, tantalum nitride, tungsten nitride, another suitable conductive material, or a combination thereof. In one example, the first electrodes **112** and the first barrier layers **1061** include the same material, such as titanium nitride. The dielectric layer **114** is conformably deposited on the first electrodes **112**. The dielectric layer **114** may include a dielectric material with a high dielectric constant (e.g. greater than or equal to 3.9). The conductive material **116** deposits on the dielectric layer **114** and fills the space between the dielectric layers **114**. The conductive material **116** includes a conductive material with a good conductivity, such as silicon germanium, high-concentration boron-doped silicon germanium, another silicon-containing conductive material, or another conductive material.

The capacitor structures **110** further include several support members **108** for preventing the collapse of the first electrodes **112** with high aspect ratio. The support members **108** may include a plurality of bottom supports **1082**, intermediate supports **1084** and top supports **1086**. The support members **108** may include, for example, silicon nitride, another suitable insulating material, or a combination thereof.

As shown in FIG. **1**A, in the peripheral region A**2**, the substrate **100** has metal contact traces **107** formed above the first insulating layer **102**. In addition, bit lines BL extend from the array region A**1** to the peripheral region A**2** and electrically connect the metal contact traces **107**.

Afterwards, the conductive material **116** is patterned, and the patterned conductive material **116'** and other components form a memory cell plate on the substrate **100**. According to some embodiments, a second insulating layer **130** (FIG. **1**E) is formed above the first insulating layer **102** and covers the memory cell plate. The second insulating layer **130** may include a single layer or multiple layers of insulating material layers.

FIG. **1**B-FIG. **1**E illustrate cross-sectional views of a method for forming a second insulating layer on a memory cell plate, in accordance with some embodiments of the present disclosure. Referring to FIG. **1**B, a dielectric material such as a first oxide material **120** is formed on the conductive material **116**. The conductive material **116** and the first oxide material **120** form in the array region A**1** and the peripheral region A**2**. The first oxide material **120** is, for example, but not limited to, silicon oxide. In various embodiments, the first oxide material **120** (the first oxide layer **120'** subsequently formed) and the second oxide layer **132** (FIG. **1**E) have different etch selectivity. By performing a deposition process, such as a physical vapor deposition (PVD) process, a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, another suitable process, or a combination of the aforementioned, to form the first oxide material **120**. In some embodiments, by performing an oxidation process to form the first oxide material **120**, for example, by thermally oxidizing the surface of the conductive material **116**.

Next, a patterning process is performed on the conductive material **116** and the first oxide material **120** to define a memory cell plate C_P and define the first oxide layer **120'** in the array region A**1**.

Referring to FIG. **1**C, a photoresist layer **122** covers a portion of the first oxide material corresponding to the array region A**1**, and exposes the remaining portion of the first oxide material **120** corresponding to the peripheral region A**2**.

Next, referring to FIG. **1**D, an etching process is performed by using the photoresist layer **122** as a mask. The portions of the first oxide material **120** and the conductive material **116** in the peripheral region A**2** are removed, and the portions of the first oxide material **120** and the conductive material **116** in the array region A**1** are remained. Next, the photoresist layer **122** is removed, for example, by an ashing process or another suitable process.

As shown in FIG. **1**D, the conductive layer **116'** and the first oxide layer **120'** are formed in the array region A**1**, and a memory cell plate C_P is defined. The memory cell plate C_P includes the contact plugs **104**, the barrier structures **106** and the capacitor structures **110**. The conductive layer **116'** includes conductor-filled portions **116**F and a covering portion **116**C. The conductor-filled portions **116**F fill up the spaces between the portions of the dielectric layer **114**. The covering portion **116**C form above the first electrodes **112** and the dielectric layer **114**. The covering portion **116**C may have a sufficient thickness to protect the underlying components of the capacitor structures **110** from being damaged in the subsequent processes. In some embodiments, the first oxide layer **120'** is conformably formed on the surface of the conductive layer **116'**, for example, formed on the top surface **116***a* and the lateral surface **116***s* of the conductive layer **116'**.

The structure illustrated in FIG. **1**D can also be obtained by using another method. In some embodiments, a patterning process can be performed on the structure illustrated in FIG. **1**A. For example, a mask is provided on the structure in FIG. **1**A, and then the portion of the conductive material **116** in the peripheral region A**2** that is not covered by the mask are removed. The portion of the conductive material **116** in the array region A**1** is remained, so as to form a memory cell plate C_P including the conductive layer **116'**. Next, a pretreatment, such as a thermal oxidation process, is performed on the conductive layer **116'** to form a first oxide layer **120'** conformably on the surface of the conductive layer **116'**.

Referring to FIG. **1**E, a second oxide layer **132** is formed over the first oxide layer **120'**. The second oxide layer **132** may include silicon oxide or another suitable oxide material. In some embodiment, the second oxide layer **132** is a tetraethyl orthosilicate (TEOS) layer. The formation method of the second oxide layer **132** may include, for example, depositing an oxide material layer in the array region A1 and the peripheral region A2 by performing PVD process, CVD process, an ALD process, spin coating, another suitable process, or a combination of the aforementioned processes. Then, the oxide material layer can be polished by performing, for example, a chemical mechanical polishing process, so as to planarize the top surfaces of the oxide material layers in the array region A**1** and the peripheral region A**2**.

In some embodiments, the second insulating layer **130** further includes a third oxide layer **134** formed above the second oxide layer **132**. The material and formation method of the third oxide layer **134** may similar to that of second oxide layer **132**.

Next, several contact holes form in the second insulating layer **130** by performing an etching process. In one embodiment, a plurality of contact holes are simultaneously formed in the second insulating layer **130** in the array region A**1** and the peripheral region A**2**.

As shown in FIG. **1**E, a pattern transfer layer **140** is formed over the second insulating layer **130**. The pattern transfer layer **140** may include a multilayer material stack, and may be formed by PVD process, CVD process, ALD process, spin coating, another suitable process, or a combination of the aforementioned processes. In some embodiments, the pattern transfer layer **140** includes a carbon-containing layer **141** and an anti-reflective layer **142** sequentially formed over the third oxide layer **134**. The carbon-containing layer **141** may include carbides, such as a diamond-like carbon film, an amorphous carbon film, a transparent carbon-containing layer. In one embodiment, the carbon-containing layer **141** is a spin-on carbon layer. The anti-reflective layer **142** may include a single layer or multiple layers of materials. The anti-reflective layer **142** includes, for example, organic polymers, carbon, silicon oxynitride, silicon oxide, another suitable material, or a combination of the aforementioned materials. In one embodiment, the anti-reflective layer **142** includes an oxygen-rich silicon oxynitride layer and a silicon oxide layer.

Next, a patterned mask layer **150** is provided over the pattern transfer layer **140**. The patterned mask layer **150** includes mask patterns **15***a* and **15***b* that are formed on the anti-reflective layer **142** in the array region A**1** and the peripheral region A**2**, respectively. The mask pattern **15***a* and **15***b* include the openings **151** and **152** respectively. The openings **151** and **152** expose parts of the top surface **142***a* of the anti-reflective layer **142** of the pattern transfer layer **140**.

Referring to FIG. **1**F, the portions of the pattern transfer layer **140** that are not covered by the patterned mask layer **150** are removed by using the patterned mask layer **150** as a mask. In one embodiment, the portions of the anti-reflective layer **142** exposed by the openings **151** and **152** and the corresponding portions of the carbon-containing layer **141** are removed by performing dry etching. Next, the pattern of the patterned mask layer **150** is transferred to the pattern transfer layer **140**, thereby forming a patterned pattern transfer layer includes a patterned anti-reflective layer (not shown) and a patterned carbon-containing layer (not shown).

Next, the patterned mask layer **150** can be removed by performing ashing process, etching process, or another suitable removing process. The second insulating layer **130** below the patterned pattern transfer layer is then etched by using the patterned pattern transfer layer as a mask to remove portions of the first insulating layer **130** for forming several contact holes. Next, the patterned pattern transfer layer is removed.

Specifically, the first contact holes **161** and the second contact hole **162** form in the second insulating layer **130** in the array region A**1** and the peripheral region A**2** respectively. The first contact holes **161** extend along the third direction D**3** and penetrate through the third oxide layer **134**, the second oxide layer **132** and the first oxide layer **120'** to reach the conductive layer **116'** of the memory cell plate C_P. That is, the bottom surfaces **161***b* of the first contact holes **161** expose the covering portion **116**C of the conductive layer **116'**.

The conductive layer **116'** functions as an etch stop layer, so that the bottom surfaces **161***b* of the first contact holes **161** and the top surface **116***a* of the conductive layer **116'** are substantially coplanar (not shown in the drawings). In some embodiments, the conductive layer **116'** is slightly etched, so that the bottom surfaces **161***b* of the first contact holes **161** are slightly lower than the top surface **116***a* of the conductive layer **116'**. After removing the patterned pattern transfer layer, the sidewalls **134***s* of the remaining third oxide layer **134'**, the sidewalls **132***s* of the remaining second oxide layer **132'** and the sidewalls **120***s* of the remaining first oxide layer **120'** expose in the first contact holes **161**.

In addition, the second contact hole **162** extends in the third direction D**3** and penetrates through the third oxide layer **134** and the second oxide layer **132** to reach the metal contact traces **107** (such as metal layer MO) in the peripheral region A**2**. The bottom surface **162***b* of the second contact hole **162** exposes the metal contact trace **107**. In addition, the first contact holes **161** and the second contact hole **162** are separated from each other in the first direction D**1**. There is a distance between the second contact hole **162** and the lateral surface of the memory cell plate C_P (i.e., the lateral surface **116***s* of the conductive layer **116'**).

Referring to FIG. **1**G, a selective etching process such as a wet etching process is performed through the first contact holes **161**. Portions of the first oxide layer **120** is removed from the first contact hole **161**, thereby forming air cavities **170** outwardly from the lower portions **161**L of the first contact holes **161**. The air cavities **170** surround and connect with the lower portions **161**L of the first contact holes **161**. The air cavities **170** expose the bottom surface **132***b* of the second oxide layer **132'** and the top surface **116***a* of the conductive layer **116'**. The etchant may include diluted hydrofluoric acid or another suitable chemical. The amount of the removed portions of the first oxide layer **120'** can be controlled by adjusting the etching time and/or the concentration of the etchant. Compared with the conventional process that only forms through holes (such as the first contact holes **161**) that extend downwardly, the embodiments of the present disclosure can further form air cavities **170** extend outwardly and recess the sidewalls of the first contact holes **161**. In the subsequent process, the first contact holes **161** and the air cavities **170** are filled with conductive material(s) to form capacitor contacts, so that the contact area between the memory cell plate C_P and each of the capacitor contact can be increased, thereby reducing the resistance of the capacitor contact. In addition, the closer the adjacent first contact holes **161** are, the easier the air cavities **170** formed around the respective contact holes are connected to each other, so that the conductive materials subsequently filled in the air cavities **170** are more easily to be connected to each other.

The first oxide layer **120'** and second oxide layer **132** have different etch selectivity. In one embodiment, the etch rate of the first oxide layer **120'** is greater than the etch rate of the second oxide layer **132**, for example, the etching ratio of the first oxide layer **120'** to the second oxide layer **132** is at least 2:1. Therefore, the second oxide layer **132** is not removed or only a small amount of the second oxide layer **132** is removed. In one embodiment, the first oxide layer **120'** is deposited by an ALD process and the material of the second oxide layer **132** is TEOS, the etch rate of the first oxide layer **120'** to the second oxide layer **132** is about 2:1. In addition, the portions of the ALD oxide can be removed by a diluted hydrofluoric acid, thereby forming the air cavities **170**.

Referring to FIG. **1**H, a conductive composite material **180** is formed above the patterned second insulating layer **130'**, to fill in the first contact holes **161**, the air cavities **170** and the second contact hole **16**. The conductive composite material **180** may include a liner layer **181** and a conductive material layer **182**. The liner layer **181** can be deposited in the first contact holes **161**, the air cavities **170** and the second contact hole **162**, and then a conductive material layer **182** is deposited over the patterned second insulating layer **130'** to fill the remaining spaces in the first contact holes **161**, the air cavities **170** and the second contact hole **162**. The liner layer **181** may include titanium, titanium nitride, tungsten nitride, tantalum, tantalum nitride, another suitable material, or a combination of the aforementioned material. The conductive material layer **182** may include tungsten, copper, another suitable metal material with good conductivity, or a combination of the aforementioned material, which have lower resistance value.

Next, the portions of the conductive material layer **182** and the liner layer **181** exceed the top surface of the patterned second insulating layer **130'** are removed by performing CMP, etch back, or a combination of the aforementioned methods. The remaining portions of the conductive composite material **180'** (FIG. **2**B) form contacts. Therefore, capacitor contacts (also referred to as first contacts) **190** are formed in the array region A**1**, and contacts (also referred to as second contacts) **194** connected to the metal contact traces **107** are formed in the peripheral region A**2**. Next, other known processes (such as formation of an interconnect structure in the peripheral region) may be performed to complete other components required for the semiconductor device.

FIG. **2**A is a schematic top view of the capacitor contacts **190** in the array region A**1**, in accordance with some embodiments of the present disclosure. FIG. **2**B is a cross-sectional view taken along sectional line **2**B-**2**B of the structure including capacitor contacts of FIG. **2**A. FIG. **2**C is a schematic diagram of a first contact portion and a second contact portion of each of the capacitor contacts, in accordance with some embodiments of the present disclosure.

Referring to FIG. **1**H and FIG. **2**A, the capacitor contacts **190** covers a portion of the top surface of the conductive layer **116'**, while the remaining portions of the top surface of the conductive layer **116'** are not covered by the capacitor contacts **190**. Therefore, when a hydrogen sintering process is performed, hydrogen is able to successfully reach and repair dangling bonds at the Si-containing material underlying of the conductive layer **116'**, thereby improving electrical performance of a semiconductor.

Referring to FIG. **2**A, FIG. **2**B and FIG. **2**C, the capacitor contacts **190** includes a first contact portion **191** and a second contact portion **192**. The first contact portion **191** includes the conductive composite material **180'** (including the liner layer **181** and the conductive material layer **182**) in the first contact hole **161**. The second contact portion **192** includes the conductive composite material **180'** in the air cavity **170**. FIG. **2**C shows a top view and a cross-sectional view of the first contact portions **191** and the second contact portions **192**. In order to show the relative positions of the first contact portion **191** and the second contact portion **192** more clearly, FIG. **2**C only depicts the first contact portion **191** and the second contact portion **192** that are respectively formed by a single-layer conductive composite material **180'**.

The capacitor contacts **190** form in the second insulating layer **130** and contact the covering portion **116**C of the conductive layer **116'** of the capacitor structure **110**. Specifically, the first contact portions **191** penetrate through the second insulating layer **130**, and electrically connect to the conductive layer **116'** of the capacitor structure **110**. The second contact portions **192** connect to the outer surface

191s of the first contact portions **191** and surrounds the lower portions **191**L of the first contact portions **191**.

Referring to FIG. **2**C, as viewed from the top of the substrate **100**, the area CA**2** surrounded by the sidewall **192**s of the second contact portion **192** is larger than the area CA**1** surrounded by the sidewall **191**s of the first contact portion **191**. The first contact portions **191** are, for example, circular or approximately circular. The second contact portions **192** surround the first contact portions **191**, wherein the diameter d**2** of the area CA**2** is larger than the diameter d**1** of the area CA**1**. In some embodiments, the first contact portions **191** and the respective second contact portions **192** are arranged in a concentric configuration.

In some other embodiments, as viewed from the top of the substrate **100**, the capacitor contacts **190** may also be configured in other shapes, such as an ellipse, a regular polygon (for example, a square or a regular hexagon) or an irregular polygon (a quadrilateral or a hexagon), or another shape.

In some embodiments, the conductive layer **116'** may be slightly etched during the formation of the first contact holes **161**. Thus, the bottom surfaces **191**b of the first contact portions **191** may be lower than the top surface **116**a of the conductive layer **116'** and the bottom surfaces **192**b of the second contact portions **192**. That is, the bottom surfaces **191**b of the first contact portions **191** are misaligned rather than coplanar with the bottom surfaces **192**b of the second contact portions **192**. In other embodiments, the conductive layer **116'** is not substantially etched, the bottom surfaces **191**b of the first contact portions **191** are substantially coplanar with the top surface **116**a of the conductive layer **116'**.

In some embodiments, the top surfaces **192**a of the second contact portions **192** are substantially coplanar with the top surface **120**a of the first oxide layer **120'**. The second contact portions **192** form over the conductive layer **116'**, so that the top surfaces **192**a of the second contact portions **192** are higher than the top surface **116**a of the conductive layer **116'**.

Referring to FIG. **2**A, FIG. **2**B and FIG. **2**C, the capacitor contacts **190** are arranged in several rows. There is a first pitch P**1** between two adjacent first contact portions **191** arranged in the same row. There is a second pitch P**2** between two adjacent first contact portions **191** arranged in different rows. The first pitch P**1** is less than the second pitch P**2**. In addition, the second contact portions **192** in the same row can connect to each other.

In other embodiments, as shown in FIG. **3**A and FIG. **3**B, the amount of the removing portions of the first oxide layer **120** can be controlled by adjusting the selective etching process (FIG. **1**G), so that adjacent air cavities **170** around the respective first contact holes **161** are not connected to each other. In one embodiment, the second contact portions **192'** of adjacent capacitor contacts **190'** are separated from each other by a distance.

In other embodiment, the capacitor contacts can be fabricated by another method. The components in FIG. **4**A-FIG. **4**F similar or identical to that in FIG. **1**A-FIG. **1**H are designated with similar or the same reference numbers, and the details are not repeated herein.

According to the method as illustrated in FIG. **1**A-FIG. **1**H, the air cavities **170** are formed in the first oxide layer **120'**. In an alternative method as illustrated in FIG. **4**A-FIG. **4**F, the air cavities **470** are formed in the conductive layer **116'**, and the second contact portions **492** of the capacitor contacts are subsequent formed after a conductive material is filled in the air cavities **470**.

Referring to FIG. **4**A, the structure shown in FIG. **1**A is provided. Then, a photoresist layer **422** is formed to cover the conductive material **116** in the array region A**1**, and expose the conductive material **116** in the peripheral region A**2**.

Referring to FIG. **4**B, the photoresist layer **422** is used as a mask, and the conductive material **116** in the peripheral region A**2** is removed. Next, the photoresist layer **422** is removed and the conductive layer **116'** is formed. A memory cell plate C_P defines in the array region A**1**.

Referring to FIG. **4**C, a second insulating layer **430** is formed over the first insulating layer **102** to cover the memory cell plate C_P in the array region A**1** and various components such as the metal contact traces **107** in the peripheral region A**2**. The second insulating layer **430** may include a first oxide layer **432** and a second oxide layer **434**. The first oxide layer **432** is conformably formed on the surfaces (i.e. the top surface **116**a and the lateral surface **116**s) of the conductive layer **116'**. The material of first oxide layer **432** may include TEOS, and the second oxide layer **434** may include silicon oxide or another suitable material. The forming method of the first oxide layer **432** and the second oxide layer **434** may be same as that of the first oxide layer **132** and the second oxide layer **134**, respectively.

Next, contact holes form in the second insulating layer **430** by using a mask and a suitable etching process. In one embodiment, contact holes simultaneously form in the second insulating layer **430** in the array region A**1** and the peripheral region A**2**

As shown in FIG. **4**C, a pattern transfer layer **440** is formed over the second insulating layer **430**. The pattern transfer layer **440** includes a carbon-containing layer **441** and an anti-reflective layer **442** sequentially formed over the second oxide layer **434**. The carbon-containing layer **441** may be a spin-on carbon layer, and the anti-reflective layer **442** may be a double-layer structure including an oxygen-rich silicon oxynitride layer and a silicon oxide layer (FIG. **4**C only shows a single-layer structure).

Next, a patterned mask layer **450** (such as a patterned photoresist layer) is provided over the pattern transfer layer **440** and exposes portion of the top surface of the pattern transfer layer **440**. The patterned mask layer **450** includes mask patterns **45**a and **45**b formed on the anti-reflection layer **442** in the array region A**1** and the peripheral region A**2**, respectively. The mask pattern **45**a **45**b includes the openings **451** and **452** which expose parts of the top surface **442**a of the anti-reflective layer **442**.

Next, the portions of the pattern transfer layer **440** that are not covered by the patterned mask layer **450** are removed. In one embodiment, the portions of the anti-reflective layer **442** exposed by the openings **451** and **452** and the corresponding portions of the carbon-containing layer **441** are removed, for example, by dry etching. Next, the pattern of the patterned mask layer **450** is transferred to the pattern transfer layer **440**, thereby forming a patterned pattern transfer layer that includes a patterned anti-reflective layer (not shown) and a patterned carbon-containing layer (not shown). Next, the patterned mask layer **450** can be removed.

Referring to FIG. **4**D, the second insulating layer **430** is etched by using the patterned pattern transfer layer as a mask to form several contact holes. Portions of the second oxide layer **434'** and the first oxide layer **432'** remain on the conductive layer **116'**. Next, the patterned pattern transfer layer is removed.

As shown in FIG. **4**D, the first contact holes **461** and the second contact hole **462** form in the second insulating layer **430** in the array region A**1** and the peripheral region A**2**. The first contact holes **461** extend along the third direction D**3** and penetrate through the second oxide layer **434** and the first oxide layer **432** to reach the conductive layer **116'** of the memory cell plate C_P. That is, the bottom surfaces **461***b* of the first contact holes **461** expose the covering portion **116**C of the conductive layer **116'**.

In addition, the conductive layer **116'** functions as an etch stop layer to stop the first contact holes **461** on the conductive layer **116'**, so that the bottom surfaces **461***b* of the first contact holes **461** and the top surface **116***a* of the conductive layer **116'** are substantially coplanar (not shown in the drawings). In some embodiments, the conductive layer **116'** is slightly etched, so as to remove small amounts of the conductive layer **116'**. Therefore, the bottom surfaces **461***b* of the first contact holes **461** are slightly lower than the top surface **116***a* of the conductive layer **116'**. After removing the patterned pattern transfer layer, the sidewalls **434***s* of the remaining second oxide layer **434'** and the sidewalls **432***s* of the remaining first oxide layer **432'** expose in the first contact holes **461**.

The second contact hole **462** extends in the third direction D**3** and penetrates through the second oxide layer **434** and the first oxide layer **432** to reach the metal contact traces **107** in the peripheral region A**2**. The bottom surface **462***b* of the second contact hole **462** exposes the metal contact trace **107**. In addition, the first contact holes **461** and the second contact hole **462** are separated from each other in the first direction D**1**. The second contact hole **462** is separated from the lateral surface of the memory cell plate C_P (i.e., the lateral surface **116***s* of the conductive layer **116'**) by a distance.

Referring to FIG. **4**E, a selective etching process is performed through the first contact holes **461** to remove portions of the conductive layer **116'**, and air cavities **470** are formed outwardly from the lower portions **461**L of the first contact holes **461**. The air cavities **470** surround and connect with the lower portions **461**L of the first contact holes **461**. The selective etching process is, for example, a wet etching process, which can isotropic remove portions of the conductive layer **116'** from the first contact hole **461**, so that the air cavities **470** expose the bottom surface **432***b* of the first oxide layer **432'** and the top surface **116***a* of the conductive layer **116'**. In some embodiments, the conductive layer **116'** includes silicon germanium, and the etchant is, for example, ammonia water or another suitable chemical. Similar to the aforementioned embodiment, when the first contact holes **461** and the air cavities **470** are filled with a conductive composite material to form the capacitor contacts, the contact area between the capacitor contacts and the memory cell plate C_P can be increased, thereby reducing the resistances of the capacitor contacts.

Referring to FIG. **4**F, a conductive composite material **480** form above the patterned second insulating layer **430'** and fill in the first contact holes **461**, the air cavities **470** and the second contact hole **462**. The conductive composite material **480** may include a liner layer **481** and a conductive material layer **482**.

Next, the excess portions of the conductive material layer **482** and the liner layer **481** above the top surface of the patterned second insulating layer **430'** are removed by, for example, CMP, etch back, or a combination of the aforementioned methods. The remaining portions of the conductive composite material **480'** (FIG. **5**B) form contacts. In some embodiments, capacitor contacts (also referred to as first contacts) **490** form in the array region A**1**, and contacts (also referred to as second contacts) **494** connected to the metal contact traces **107** form in the peripheral region A**2**. The capacitor contacts **490** includes a first contact portion **491** and a second contact portion **492** that surrounds the lower portion **491**L of the first contact portion **491**.

FIG. **5**A is a schematic top view of capacitor contacts **490** in the array region A**1**. FIG. **5**B is a cross-sectional view taken along sectional line **5**B-**5**B of the structure including capacitor contacts of FIG. **5**A. Although the second contact portions **492** of the capacitor contacts **490** shown in FIG. **4**E, FIG. **4**F, FIG. **5**A and FIG. **5**B are connected to each other, the present disclosure is not limited thereto. In some other embodiments, the amount of the removing portions of the conductive layer **116'** can be controlled by adjusting the selective etching process (FIG. **4**E), so that adjacent air cavities **470** around the respective first contact holes **461** are not connected to each other.

Referring to FIG. **4**F and FIG. **5**A, only parts of the top surface of the conductive layer **116'** are covered by the capacitor contacts **490**. Therefore, when H_2 sintering process is performed, hydrogen is able to successfully reach and repair dangling bonds at the Si-containing material underlying of the conductive layer **116'**.

Referring to FIG. **4**E, FIG. **4**F, FIG. **5**A and FIG. **5**B, the capacitor contacts **490** includes a first contact portion **491** and a second contact portion **492**. The first contact portion **491** includes a portion of the conductive composite material **480'** (including the liner layer **481** and the conductive material layer **482**) in the first contact hole **461**. The second contact portion **492** includes the other portion of the conductive composite material **480'** in the air cavity **470**.

The capacitor contacts **490** form between the second insulating layer **430** and a portion of the conductive layer **116'**. The first contact portions **491** penetrate through the second insulating layer **430** and a portion of the conductive layer **116'**. Therefore, the capacitor contacts **490** contact and electrically connect the covering portion **116**C of the conductive layer **116'** of the capacitor structure **110**. The second contact portions **492** of the capacitor contacts **490** connect to the outer surface **491***s* of the first contact portions **491** and surrounds the lower portions **491**L of the respective first contact portions **491**.

Referring to FIG. **5**A, as viewed from the top of the substrate **100**, the area surrounded by the sidewall of the second contact portion **492** is greater than the area surrounded by the sidewall of the first contact portion **491**. The first contact portions **491** and the second contact portions **492** are, for example, circular or another shape. In some embodiments, the second contact portions **492** surround the first contact portions **491**. In some embodiments, the first contact portions **491** and the second contact portions **492** are arranged in a concentric configuration.

In some embodiments, an isotropic etching process is performed on the conductive layer **116'** exposed at the lower portions of the first contact holes **461** to form the air cavities **470**. Therefore, after the capacitor contacts **490** are formed, the bottom surfaces **491***b* of the first contact portions **491** are lower than the top surface **116***a* of the conductive layer **116'** and are substantially coplanar with the bottom surfaces **492***b* of the second contact portions **492**. That is, the second contact portions **492** are formed in the conductive layer **116'**. In some embodiments, the top surfaces **492***a* of the second contact portions **492** are substantially coplanar with the top surface **116***a* of the conductive layer **116'**.

According to the aforementioned descriptions, the semiconductor device and the manufacturing methods thereof are provided. The capacitor contacts of the embodiment do not fully cover the memory cell plate. Therefore, when a hydrogen sintering process is performed, hydrogen is able to reach the second insulating layer and a portion of the conductive layer (including such as silicon germanium material) that is in direct contact with the second insulating layer. Thus, hydrogen is able to repair dangling bonds at the Si-containing material, thereby repairing the process defects and improving the electrical performance of a semiconductor device. For example, the threshold voltage of the semiconductor device can be reduced. In some embodiments, among the capacitor contacts that are connected to the capacitor structure, the first contact portions of the capacitor contacts penetrate through the insulating layer and are in contact with the underlying conductive layer, while the second contact portions surrounds the bottoms of the respective first contact portions. Therefore, the contact area between the capacitor contacts and the conductive layer of the capacitor structure can be increased, so as to reduce the high resistance of the capacitor contacts caused by the material of the conductive layer (e.g., silicon germanium material). In addition, the methods of manufacturing the semiconductor device, in accordance with some embodiments of the present disclosure, do not include complicated and expensive manufacturing processes, which save production time for fabricating the semiconductor device and do not increase the manufacturing cost.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device, comprising:
- a first insulating layer over a substrate;
- a contact plug in the first insulating layer and in contact with a surface of the substrate;
- a capacitor structure above the contact plug, wherein the capacitor structure includes a conductive layer over the first insulating layer;
- a second insulating layer over the first insulating layer, wherein the second insulating layer covers the capacitor structure; and
- a capacitor contact over the capacitor structure, wherein the capacitor contact comprises:
- a first contact portion penetrating through the second insulating layer and extending into the conductive layer of the capacitor structure, wherein the first contact portion comprises a lower portion in contact with the conductive layer of the capacitor structure; and
- a second contact portion connecting an outer surface of the first contact portion and surrounding the lower portion of the first contact portion, wherein the second contact portion extends outwardly from the lower portion of the first contact portion.

2. The semiconductor device as claimed in claim **1**, wherein an area surrounded by sidewalls of the second contact portion is greater than an area surrounded by sidewalls of the first contact portion as viewed from a top side of the substrate.

3. The semiconductor device as claimed in claim **1**, wherein a bottom surface of the first contact portion is lower than a top surface of the conductive layer of the capacitor structure.

4. The semiconductor device as claimed in claim **1**, wherein the second contact portion is formed on the conductive layer of the capacitor structure.

5. The semiconductor device as claimed in claim **4**, wherein the second insulating layer comprises:
- a first oxide layer on the conductive layer; and
- a second oxide layer on the first oxide layer,
- wherein there is an etch selectivity between the first oxide layer and the second oxide layer.

6. The semiconductor device as claimed in claim **5**, wherein a top surface of the second contact portion is level with a top surface of the first oxide layer of the second insulating layer.

7. The semiconductor device as claimed in claim **1**, wherein the second contact portion is positioned within the conductive layer of the capacitor structure.

8. The semiconductor device as claimed in claim **1**, wherein a top surface of the second contact portion is level with a top surface of the conductive layer of the capacitor structure.

9. The semiconductor device as claimed in claim **1**, wherein a bottom surface of the first contact portion is misaligned with a bottom surface of the second contact portion.

10. The semiconductor device as claimed in claim **1**, wherein a bottom surface of the first contact portion is coplanar with a bottom surface of the second contact portion.

11. The semiconductor device as claimed in claim **1**, comprising a plurality of capacitor contacts, wherein adjacent second contact portions of the capacitor contacts are separated from each other by a distance.

12. The semiconductor device as claimed in claim **1**, comprising a plurality of capacitor contacts, wherein adjacent second contact portions of the capacitor contacts are connected to each other.

13. The semiconductor device as claimed in claim **1**, comprising:
- a plurality of capacitor contacts arranged in rows, wherein the first contact portions of two adjacent capacitor contacts arranged in the same row are spaced apart by a first pitch, the first contact portions of two adjacent capacitor contacts arranged in different rows are spaced apart by a second pitch, and the first pitch is less than the second pitch,
- wherein the second contact portions of the capacitor contacts that are arranged in the same row are connected to each other.

14. The semiconductor device as claimed in claim **1**, wherein the capacitor contact is referred to as a first contact in an array region of the substrate, and the semiconductor device further comprises:
- a second contact in a peripheral region of the substrate, wherein the second contact penetrates through the second insulating layer and is electrically connected to a metal contact trace in the peripheral region.

15. A method of manufacturing a semiconductor device, comprising:
- providing a substrate and forming a memory cell plate over the substrate, wherein the memory cell plate comprises:
- a first insulating layer over the substrate;

a contact plug in the first insulating layer and in contact with a surface of the substrate; and

a capacitor structure above the contact plug, wherein the capacitor structure includes a conductive layer over the first insulating layer;

forming a second insulating layer over the first insulating layer, wherein the second insulating layer covers the memory cell plate; and

forming a capacitor contact over the capacitor structure, wherein the capacitor contact comprises:

a first contact portion penetrating through the second insulating layer and extending into the conductive layer of the capacitor structure, wherein the first contact portion comprises a lower portion in contact with the conductive layer of the capacitor structure; and

a second contact portion connecting an outer surface of the first contact portion and surrounding the lower portion of the first contact portion, wherein the second contact portion extends outwardly from the lower portion of the first contact portion.

16. The method of manufacturing a semiconductor device as claimed in claim **15**, wherein forming the second insulating layer comprises:

forming a first oxide layer on the memory cell plate; and

forming a second oxide layer on the first oxide layer,

wherein there is an etch selectivity between the first oxide layer and the second oxide layer.

17. The method of manufacturing a semiconductor device as claimed in claim **16**, further comprising simultaneously patterning a first oxide material and a conductive material to form the first oxide layer and the conductive layer, respectively.

18. The method of manufacturing a semiconductor device as claimed in claim **16**, wherein forming the capacitor contact comprises:

forming a pattern transfer layer over the conductive layer of the capacitor structure;

forming a patterned mask layer over the pattern transfer layer;

transferring a pattern of the patterned mask layer to the pattern transfer layer;

removing portions of the second insulating layer to form first contact holes in the second insulating layer by using the patterned pattern transfer layer as a mask, wherein the first contact holes penetrate the second oxide layer and the first oxide layer, and bottom surfaces of the first contact holes expose the conductive layer;

performing a selective etching process through the first contact hole to remove portions of the first oxide layer, so as to form an air cavity from the lower portion of the first contact hole, wherein the air cavity surrounds and communicates with the lower portion of the first contact hole; and

filling a conductive composite material into the first contact hole and the air cavity.

19. The method of manufacturing a semiconductor device as claimed in claim **15**, wherein the second insulating layer is in contact with the conductive layer of the capacitor structure.

20. The method of manufacturing a semiconductor device as claimed in claim **19**, wherein forming the capacitor contact comprises:

forming a pattern transfer layer over the conductive layer of the capacitor structure;

forming a patterned mask layer over the pattern transfer layer;

transferring a pattern of the patterned mask layer to the pattern transfer layer;

removing portions of the second insulating layer to form first contact holes in the second insulating layer by using the patterned pattern transfer layer as a mask, wherein bottom surfaces of the first contact holes expose the conductive layer;

performing a selective etching process through the first contact hole to remove portions of the first oxide layer, so as to remove a portion of the conductive layer and form an air cavity from the lower portion of the first contact hole, wherein the air cavity surrounds the lower portion of the first contact hole, and the air cavity communicates with the first contact hole; and

filling a conductive composite material into the first contact hole and the air cavity.

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